

Use-Case Model - Dormify Dormitory Management System

Performed by: Trần Huỳnh Mạnh Đạt | Reviewed by: Đào Duy Anh | Edited by: Đào Duy Anh

1. Actors

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Actor	Description
Applicant / Guest	A person who has not yet become a dormitory resident and submits preliminary information to apply for accommodation.
Student	A dormitory resident who manages personal information, room applications, contracts, invoices, residence declarations, repair requests, feedback, and conduct information.
System Admin	Manages system accounts, roles, permissions, audit logs, and the administration dashboard.
Dormitory Manager	Performs dormitory operations, including room, student, contract, finance, residence, maintenance, feedback, and conduct management.
Maintenance Staff	Receives assigned maintenance jobs and updates repair progress and results.
Google OAuth Provider	External authentication provider used for Google login.
Email / SMS Service	Sends contact notifications and password-reset OTP codes.
Payment Gateway	Processes invoice payments through supported payment methods such as banks, VNPay, MoMo, or ZaloPay.
Scheduled Trigger	Initiates time-based automated operations such as marking overdue invoices and sending debt reminders.

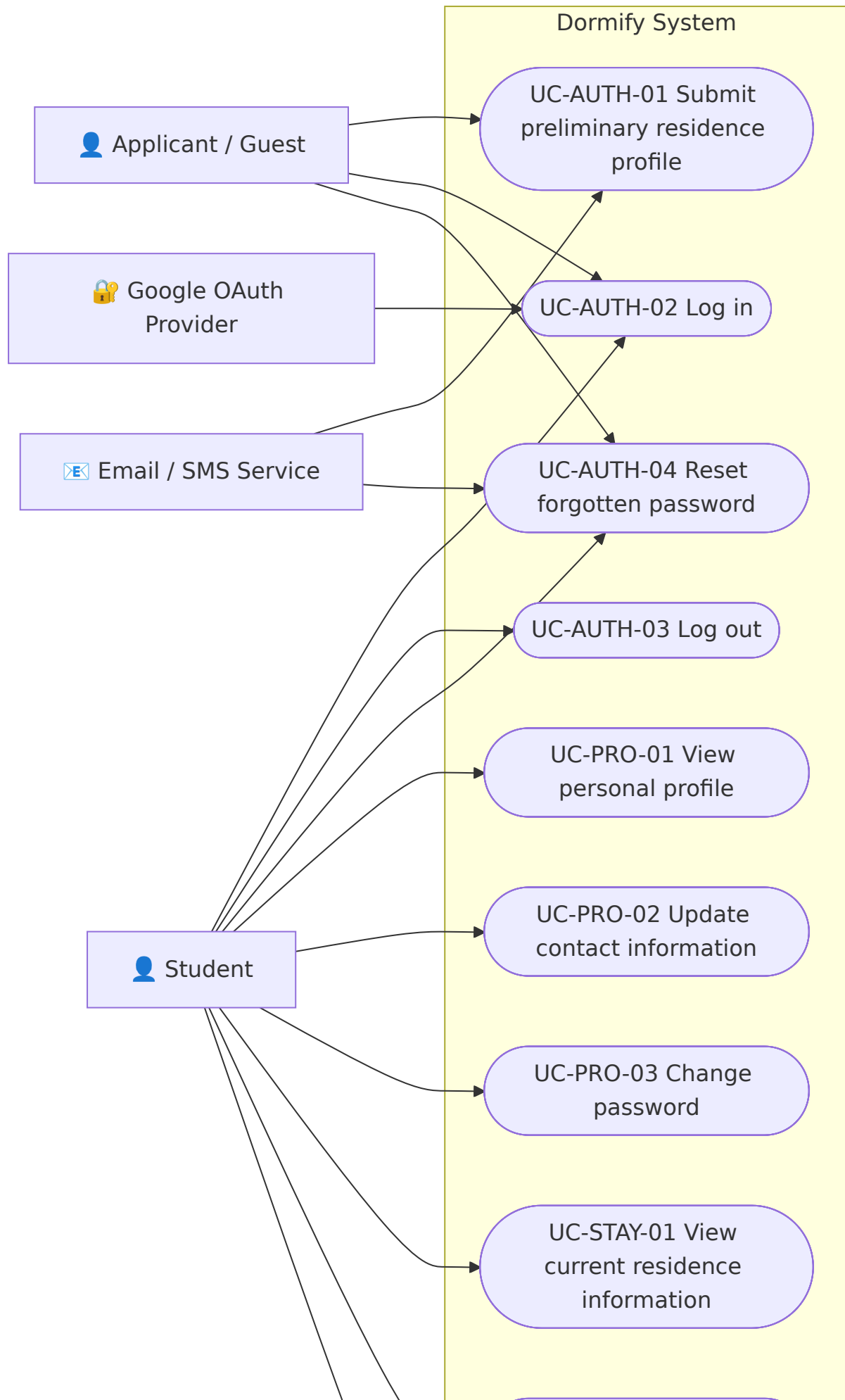
The **System Admin** and **Dormitory Manager** are separate roles. They are not generalized into a common Manager actor because they have different responsibilities and access permissions.

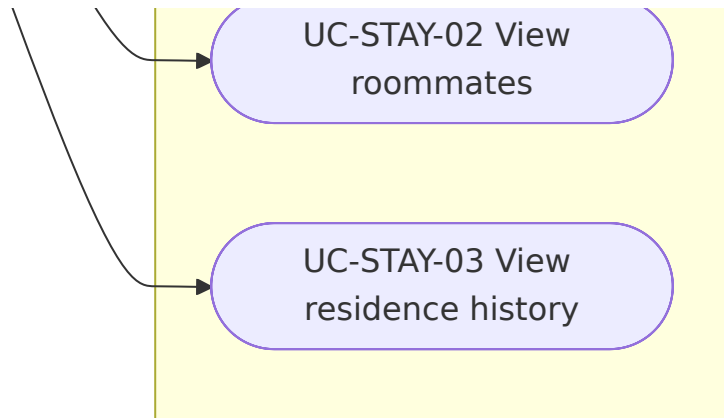
Modeling note (Floor Manager): This diagram set presents former Floor Manager responsibilities as folded into the Dormitory Manager actor, for a simpler and more readable diagram. This is a **presentation simplification only** — the backend's RBAC model still defines `FLOOR_MANAGER` as a real, distinct role (`users.schema.ts` → `USER_ROLES`) and several endpoints (transfers, absences, bookings review, contracts list, violations lookup) still guard on it explicitly via `@Roles( . . . )`. No code change to remove the role has been made. If the team decides to actually retire `FLOOR_MANAGER`, that requires a separate backend change (remove it from the enum and every `@Roles()` list it appears in) — see `system_plan.md` § 3 for where it is currently used.

2. Authentication and Personal Profile

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2.1 Use-Case Diagram





2.2 Use-Case Summary

ID	Use case	Primary actor
UC-AUTH-01	Submit preliminary residence profile	Applicant / Guest
UC-AUTH-02	Log in	Applicant / Guest, Student
UC-AUTH-03	Log out	Student
UC-AUTH-04	Reset forgotten password	Applicant / Guest, Student
UC-PRO-01	View personal profile	Student
UC-PRO-02	Update contact information	Student
UC-PRO-03	Change password	Student
UC-STAY-01	View current residence information	Student
UC-STAY-02	View roommates	Student
UC-STAY-03	View residence history	Student

Notes

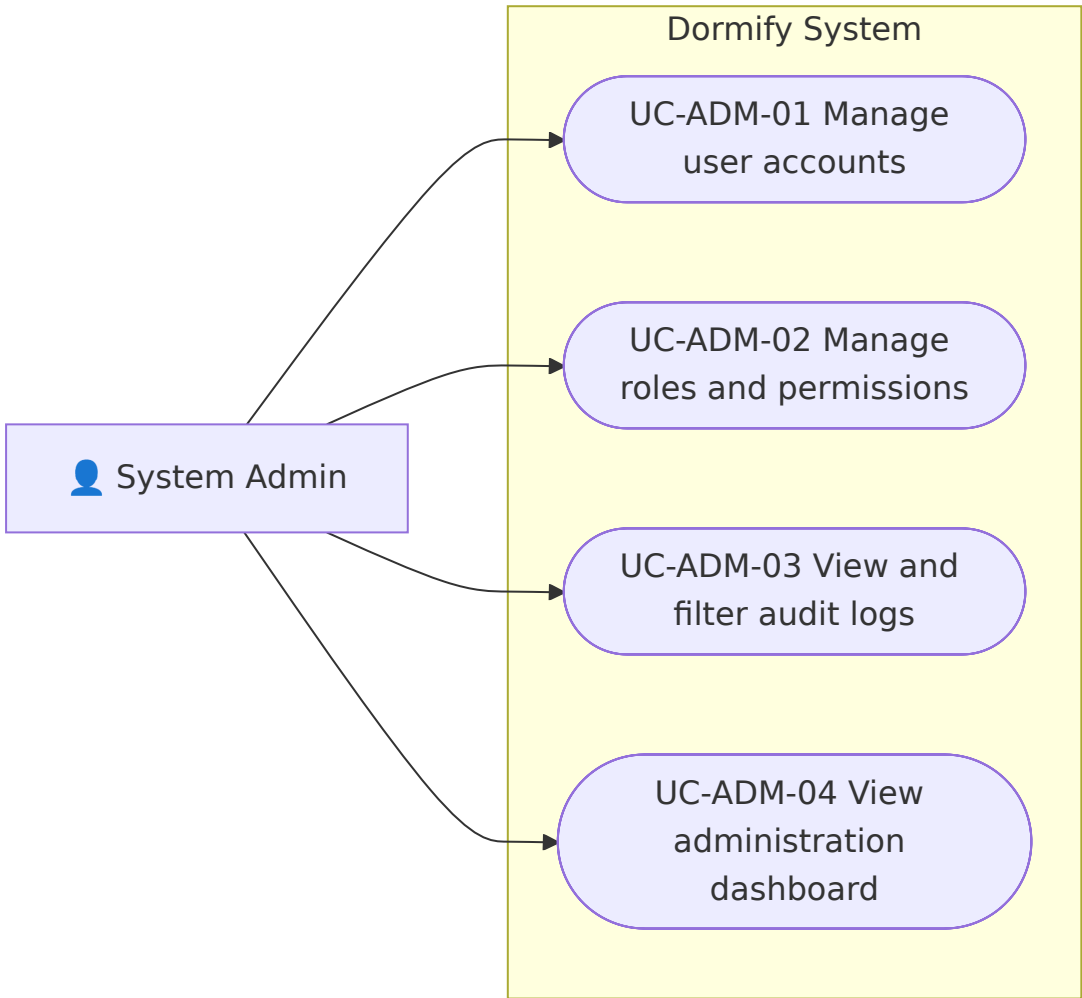
- UC - AUTH - 02 supports login using email, citizen identification number, or Google authentication.
- Google login is treated as an alternative flow of UC - AUTH - 02 , not as a separate use case.
- UC - AUTH - 04 sends an OTP through the configured email or SMS service.
- Students may update contact information only.
- Identity-related information such as full name, date of birth, and identification number may only be updated by authorized management roles.

- Current residence information includes building, floor, room, and bed.

3. System Administration

Performed by: Trần Huỳnh Mạnh Đạt | **Reviewed by:** Đào Duy Anh | **Edited by:** Đào Duy Anh

3.1 Use-Case Diagram



3.2 Use-Case Summary

ID	Use case	Main operations
UC-ADM-01	Manage user accounts	View, lock, unlock, deactivate, or delete user accounts
UC-ADM-02	Manage roles and permissions	Create roles, assign permissions, assign roles, and revoke access

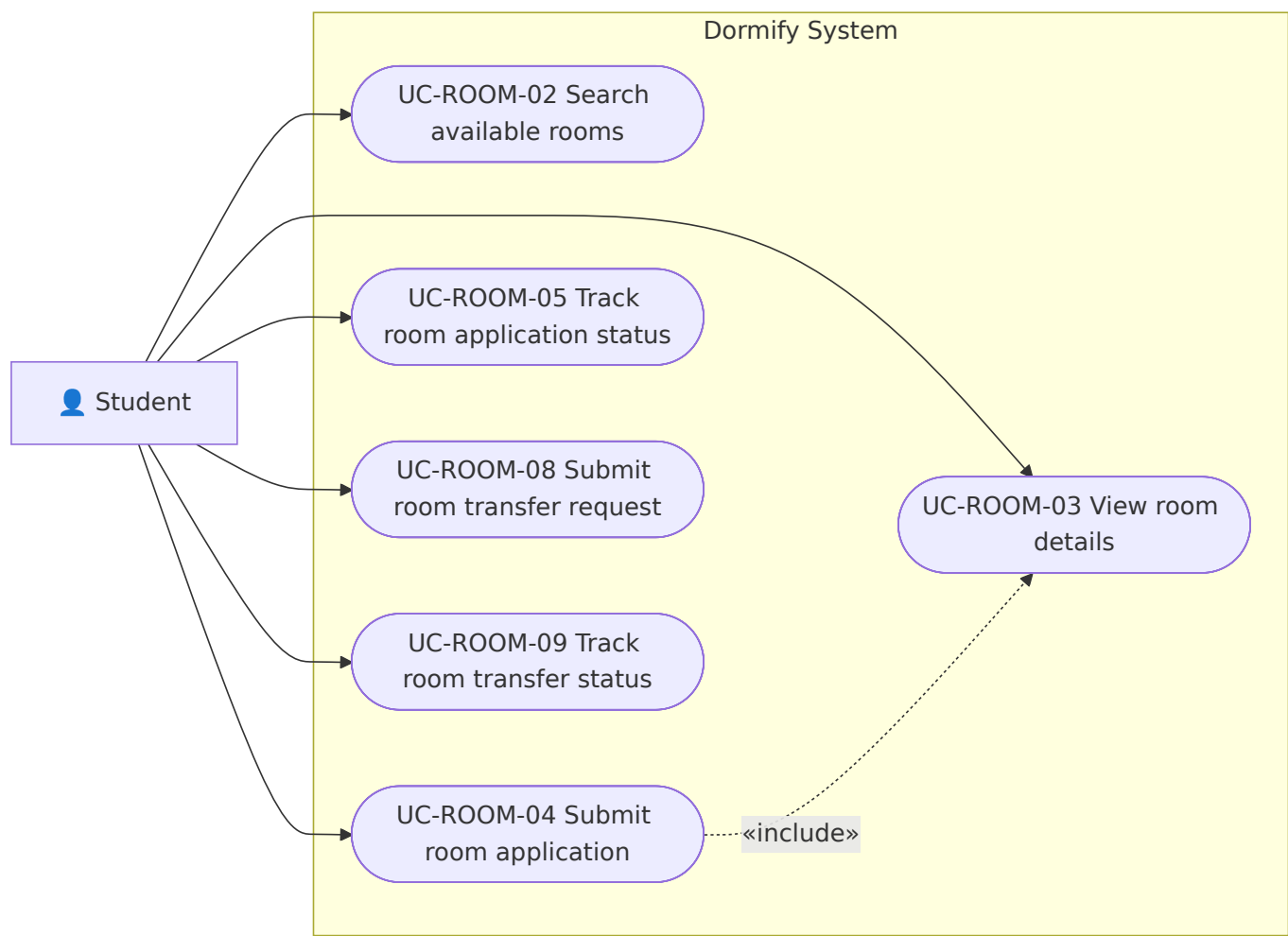
UC-ADM-03	View and filter audit logs	Review actions, users, timestamps, changed data, and used functions
UC-ADM-04	View administration dashboard	View system statistics, warnings, and operational summaries

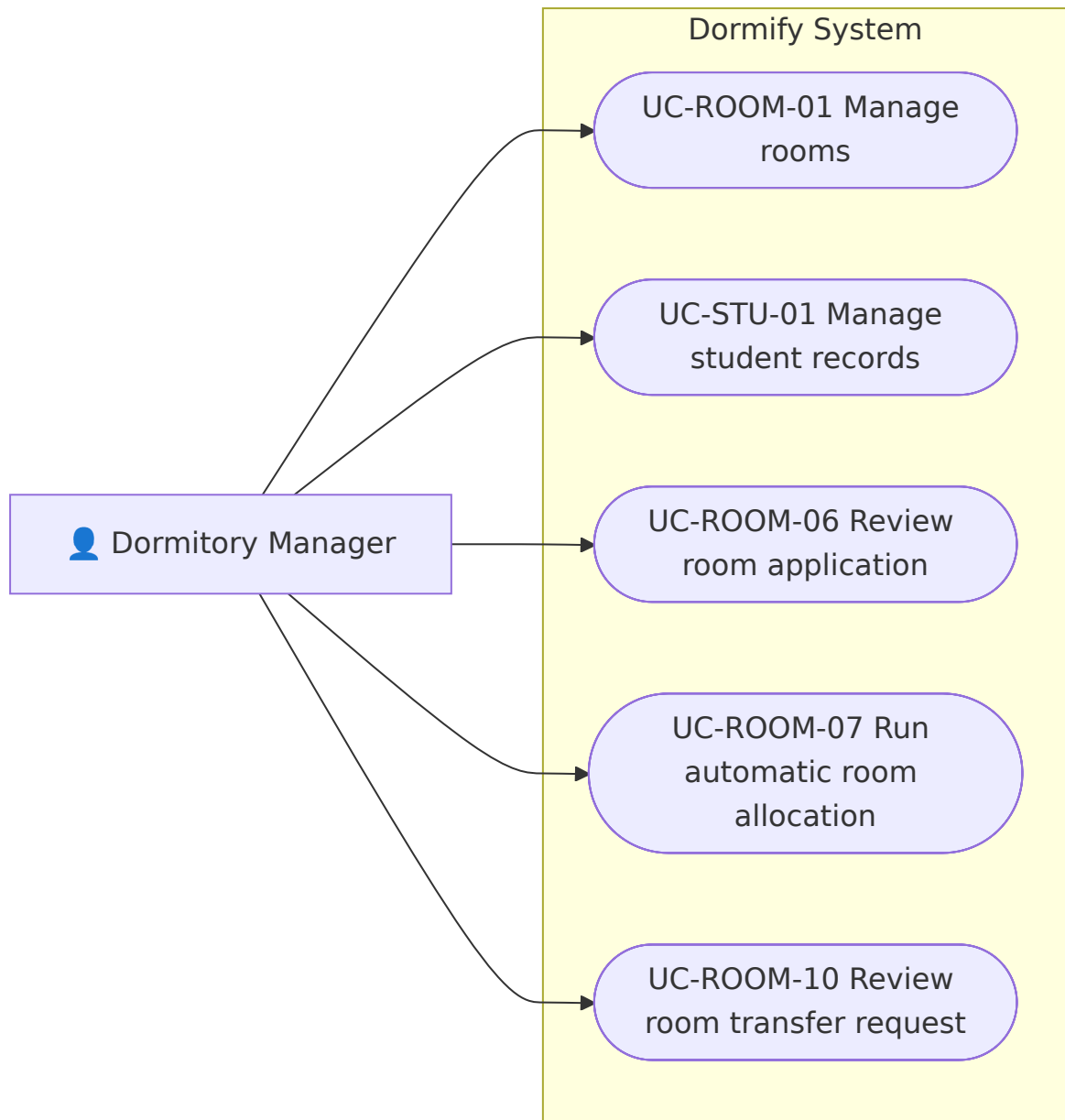
FR08 - Back up and restore data is not included because it has been removed from the current project scope.

4. Room and Student Management

Performed by: Trần Huỳnh Mạnh Đạt | Reviewed by: Đào Duy Anh | Edited by: Đào Duy Anh

4.1 Use-Case Diagram





4.2 Use-Case Summary

ID	Use case	Main operations
UC-ROOM-01	Manage rooms	Add, update, remove, view, and mark rooms as under maintenance
UC-STU-01	Manage student records	View, edit, filter, and review room-rental history
UC-ROOM-02	Search available rooms	Search by room type, building, price, and available beds
UC-ROOM-03	View room details	View room information, price, capacity, and available beds

UC-ROOM-04	Submit room application	Submit a request to rent a selected room
UC-ROOM-05	Track room application status	View pending, approved, or rejected status
UC-ROOM-06	Review room application	Check the applicant profile and approve or reject the application
UC-ROOM-07	Run automatic room allocation	Allocate students based on preferences, gender, capacity, and availability
UC-ROOM-08	Submit room transfer request	Request a transfer to another room
UC-ROOM-09	Track room transfer status	View the processing status of a transfer request
UC-ROOM-10	Review room transfer request	Approve or reject a transfer and record transfer history

Reassigned Floor Manager Responsibilities (Room & Student Management)

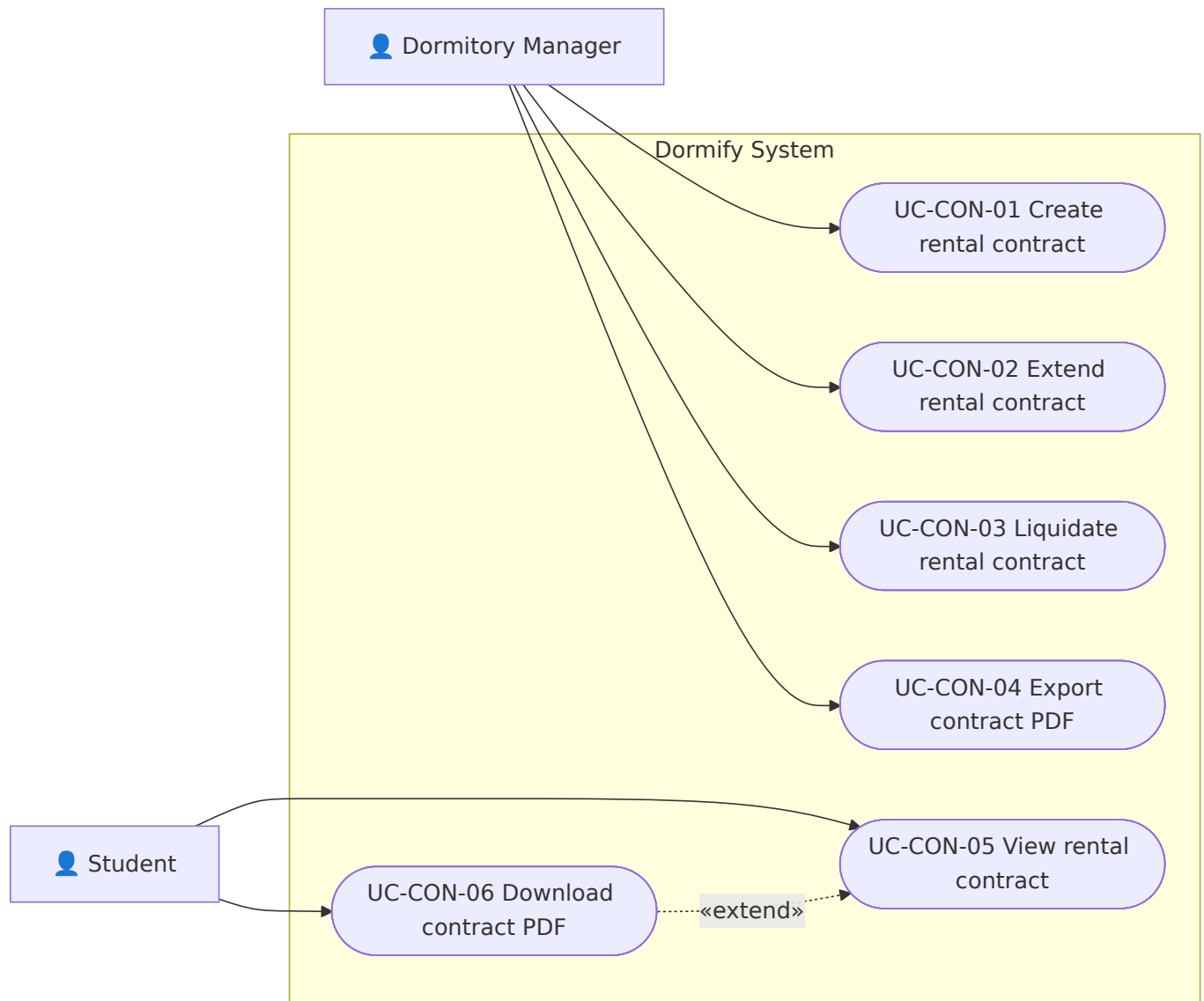
Because Floor Manager responsibilities are folded into the Dormitory Manager actor in this diagram set (see the modeling note in § 1):

- Viewing students by floor is handled in UC-STU-01 .
- Viewing residence information is handled in UC-STU-01 .
- Monitoring room conditions by floor is handled in UC-ROOM-01 .
- Dormitory Manager screens should support filtering by building and floor.

5. Contract Management

Performed by: Trần Huỳnh Mạnh Đạt | Reviewed by: Đào Duy Anh | Edited by: Đào Duy Anh

5.1 Use-Case Diagram



5.2 Use-Case Summary

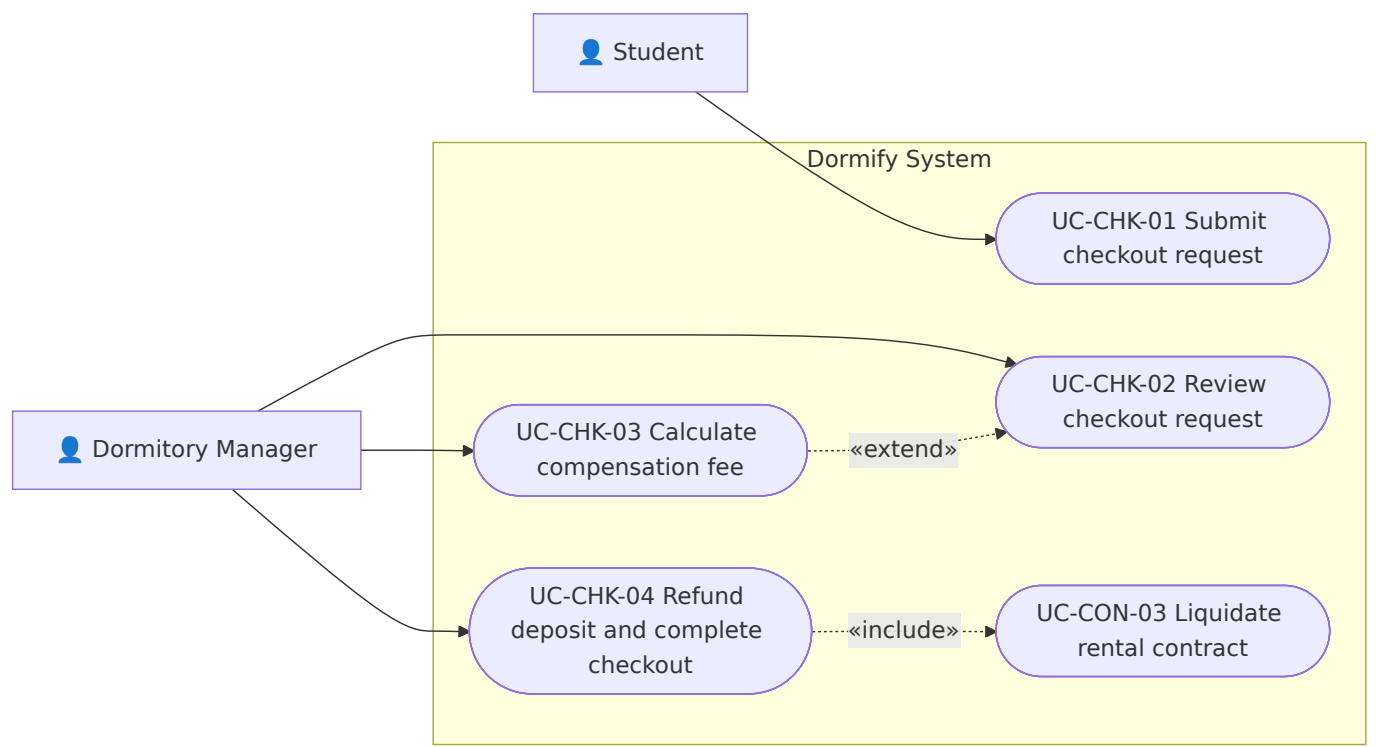
ID	Use case	Primary actor
UC-CON-01	Create rental contract	Dormitory Manager
UC-CON-02	Extend rental contract	Dormitory Manager
UC-CON-03	Liquidate rental contract	Dormitory Manager
UC-CON-04	Export contract PDF	Dormitory Manager
UC-CON-05	View rental contract	Student
UC-CON-06	Download contract PDF	Student

A rental contract may be created after a room application or automatic room allocation has been approved.

6. Checkout and Deposit Refund

Performed by: Trần Huỳnh Mạnh Đạt | **Reviewed by:** Đào Duy Anh | **Edited by:** Đào Duy Anh

6.1 Use-Case Diagram



6.2 Use-Case Summary

ID	Use case	Description
UC-CHK-01	Submit checkout request	Student requests to leave the assigned room
UC-CHK-02	Review checkout request	Manager reviews and processes the request
UC-CHK-03	Calculate compensation fee	Manager enters damaged or missing items and calculates deductions
UC-CHK-04	Refund deposit and complete checkout	System calculates the remaining deposit and closes the stay

Scope Note

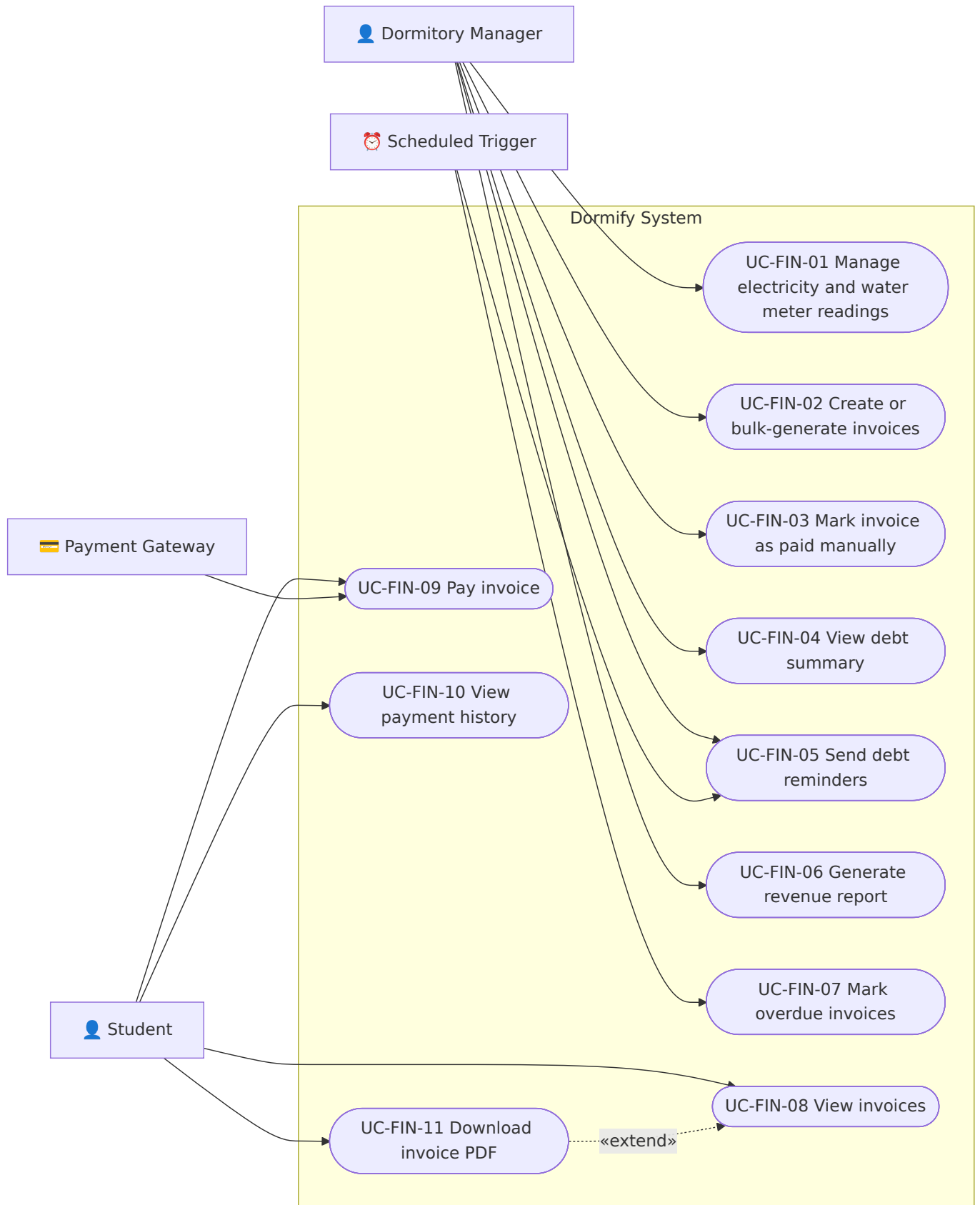
FR19 - Inspect assets has been removed as a standalone use case.

When necessary, damaged or missing items may be entered manually during checkout processing. Compensation calculation only occurs when deductions are recorded.

7. Finance and Meter Management

Performed by: Trần Huỳnh Mạnh Đạt | **Reviewed by:** Đào Duy Anh | **Edited by:** Đào Duy Anh

7.1 Use-Case Diagram



7.2 Use-Case Summary

ID	Use case	Main operations
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UC-FIN-01	Manage electricity and water meter readings	Enter opening and closing readings, upload meter photos, and view consumption history
UC-FIN-02	Create or bulk-generate invoices	Create electricity, water, accommodation, or repair invoices
UC-FIN-03	Mark invoice as paid manually	Confirm offline or manually verified payments
UC-FIN-04	View debt summary	View unpaid and overdue amounts by student or room
UC-FIN-05	Send debt reminders	Send manual or scheduled reminders
UC-FIN-06	Generate revenue report	Generate monthly, quarterly, or yearly revenue reports
UC-FIN-07	Mark overdue invoices	Automatically change eligible pending invoices to overdue
UC-FIN-08	View invoices	View invoice list and invoice details
UC-FIN-09	Pay invoice	Pay using a bank or supported electronic payment provider
UC-FIN-10	View payment history	Review successful, pending, and failed payments
UC-FIN-11	Download invoice PDF	Export an invoice as a PDF file

Reassigned Floor Manager Responsibilities (Finance & Meter Management)

The following Floor Manager functions are folded into the Dormitory Manager actor in this diagram set:

- Enter electricity opening and closing readings.
- Enter water opening and closing readings.
- Upload meter photographs.

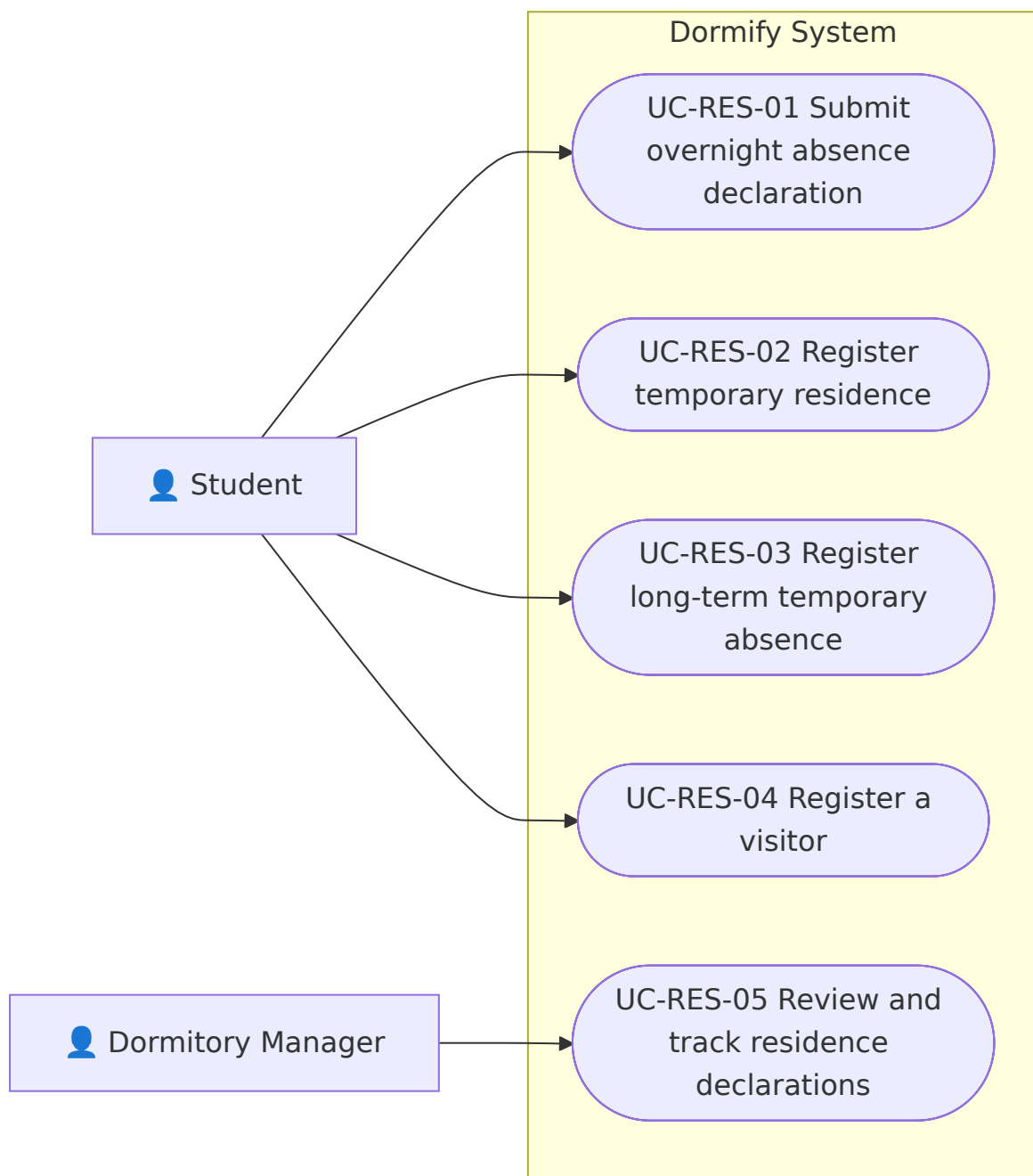
- Review electricity and water consumption history.

The payment methods supported by UC-FIN-09 are alternative flows within the same use case, not separate use cases.

8. Residence Management

Performed by: Trần Huỳnh Mạnh Đạt | **Reviewed by:** Đào Duy Anh | **Edited by:** Đào Duy Anh

8.1 Use-Case Diagram



8.2 Use-Case Summary

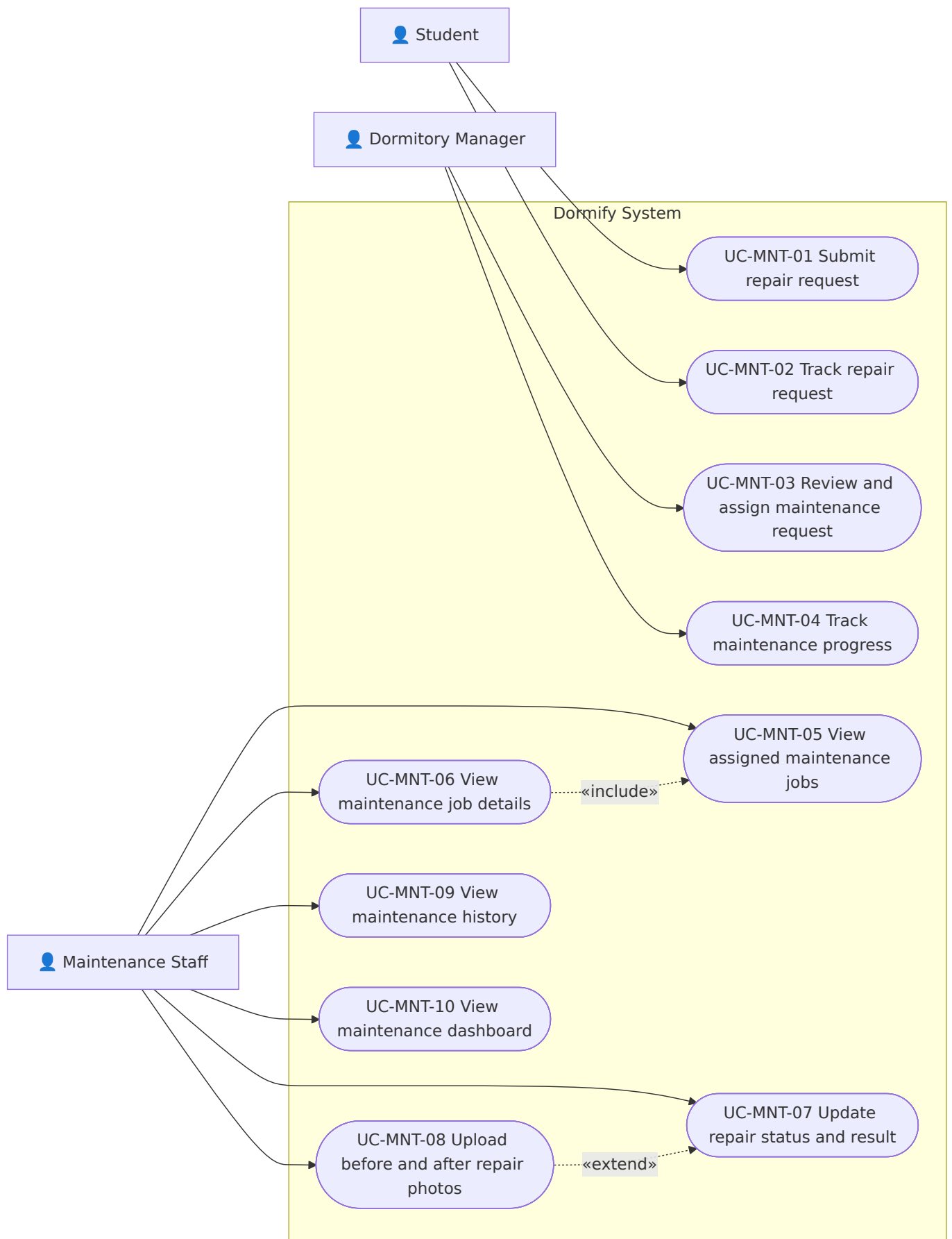
ID	Use case	Description
UC-RES-01	Submit overnight absence declaration	Student reports an overnight absence
UC-RES-02	Register temporary residence	Student submits temporary residence information
UC-RES-03	Register long-term temporary absence	Student reports a long-term absence
UC-RES-04	Register a visitor	Student registers visitor information and visit time
UC-RES-05	Review and track residence declarations	Manager views and processes residence-related declarations

The Dormitory Manager actor assumes all residence-monitoring responsibilities that this diagram set folds in from Floor Manager (see the modeling note in § 1).

9. Maintenance Management

Performed by: Trần Huỳnh Mạnh Đạt | **Reviewed by:** Đào Duy Anh | **Edited by:** Đào Duy Anh

9.1 Use-Case Diagram



9.2 Use-Case Summary

ID	Use case	Description
UC-MNT-01	Submit repair request	Student reports an incident and may attach photographs
UC-MNT-02	Track repair request	Student views request status and repair results
UC-MNT-03	Review and assign maintenance request	Manager reviews and assigns a request to staff
UC-MNT-04	Track maintenance progress	Manager monitors pending, in-progress, and completed requests
UC-MNT-05	View assigned maintenance jobs	Staff views jobs assigned to them
UC-MNT-06	View maintenance job details	Staff views issue, location, priority, and attachments
UC-MNT-07	Update repair status and result	Staff changes status and records the repair result
UC-MNT-08	Upload before and after repair photos	Staff provides visual evidence of repair work
UC-MNT-09	View maintenance history	Staff reviews previously processed maintenance work
UC-MNT-10	View maintenance dashboard	Staff views workload and job-status statistics

Valid repair statuses include:

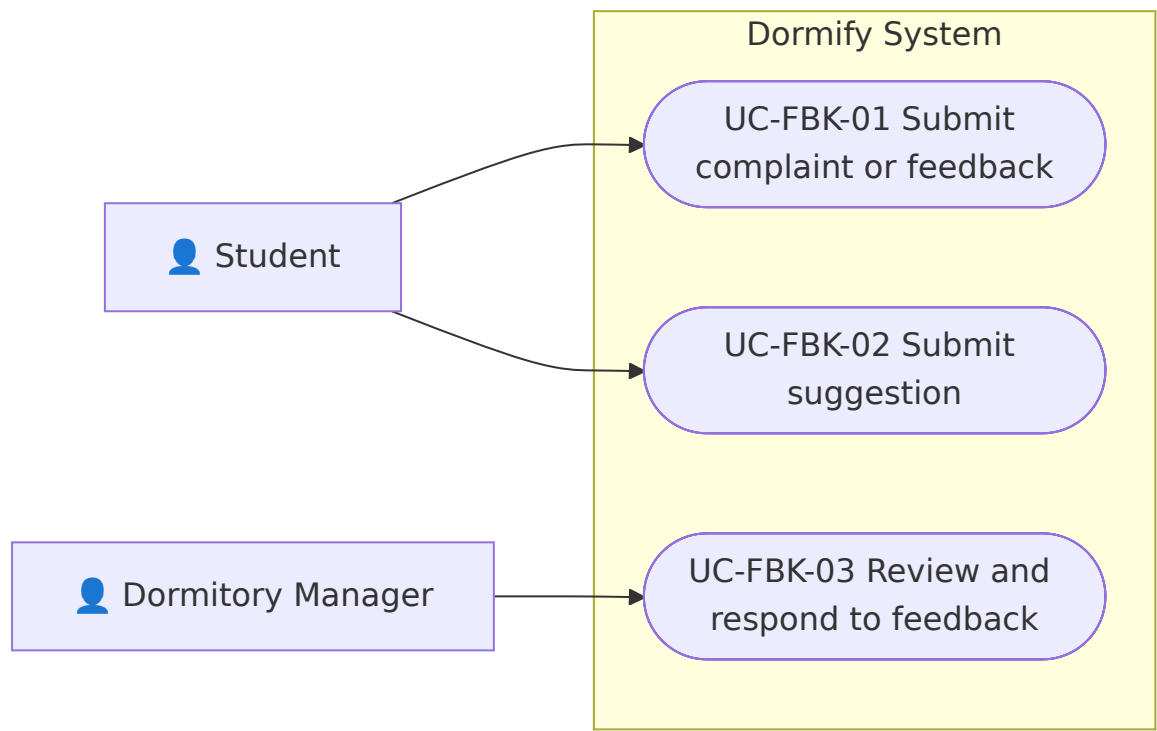
- Pending.
- In progress.
- Completed.

The assignment action may contact an internal employee or a third-party maintenance provider, depending on the final implementation.

10. Feedback and Suggestions

Performed by: Trần Huỳnh Mạnh Đạt | Reviewed by: Đào Duy Anh | Edited by: Đào Duy Anh

10.1 Use-Case Diagram



10.2 Use-Case Summary

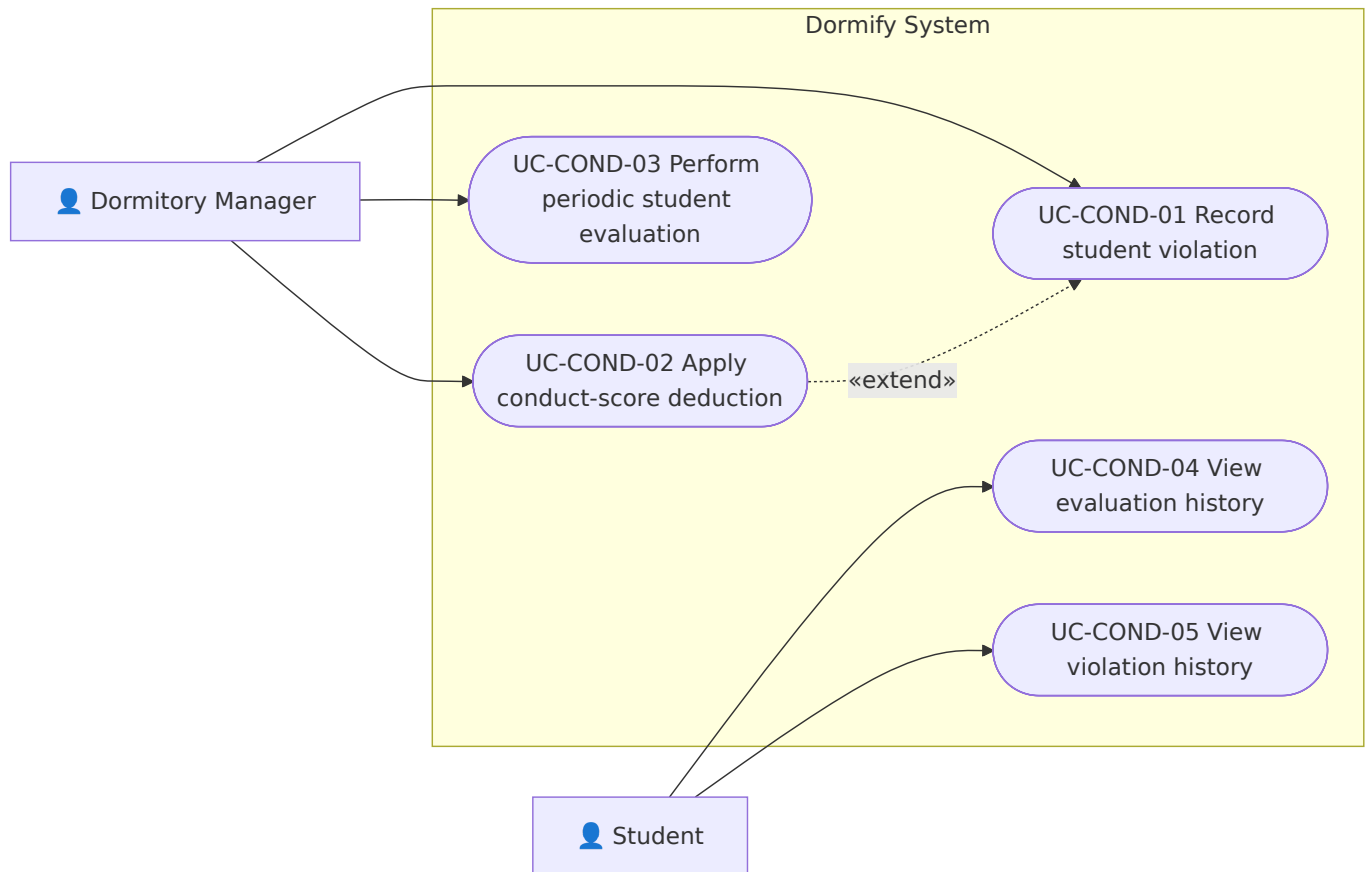
ID	Use case	Primary actor
UC-FBK-01	Submit complaint or feedback	Student
UC-FBK-02	Submit suggestion	Student
UC-FBK-03	Review and respond to feedback	Dormitory Manager

The Dormitory Manager actor assumes the feedback-handling responsibility that this diagram set folds in from Floor Manager (see the modeling note in § 1).

11. Conduct and Student Evaluation

Performed by: Trần Huỳnh Mạnh Đạt | Reviewed by: Đào Duy Anh | Edited by: Đào Duy Anh

11.1 Use-Case Diagram



11.2 Use-Case Summary

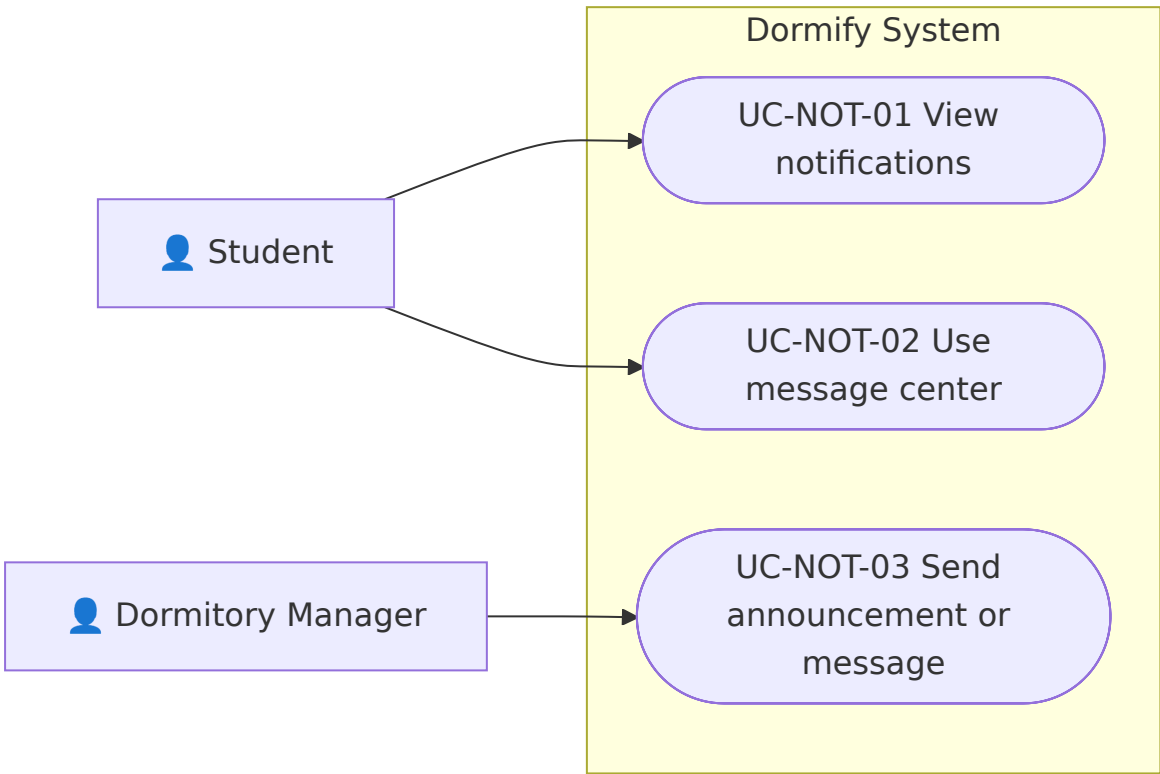
ID	Use case	Description
UC-COND-01	Record student violation	Manager creates a violation record
UC-COND-02	Apply conduct-score deduction	System applies a deduction based on configured dormitory rules
UC-COND-03	Perform periodic student evaluation	Manager records a periodic conduct evaluation
UC-COND-04	View evaluation history	Student views previous conduct evaluations
UC-COND-05	View violation history	Student views recorded violations and related penalties

UC-COND-02 extends UC-COND-01 because not every recorded incident must necessarily result in a conduct-score deduction.

12. Notifications and Message Center

Performed by: Trần Huỳnh Mạnh Đạt | Reviewed by: Đào Duy Anh | Edited by: Đào Duy Anh

12.1 Use-Case Diagram



12.2 Use-Case Summary

ID	Use case	Description
UC-NOT-01	View notifications	Student views system-generated and management notifications
UC-NOT-02	Use message center	Student views received messages and communication history
UC-NOT-03	Send announcement or message	Manager sends information to selected students, rooms, floors, or buildings

Automatic notifications are treated as postconditions of related use cases.

Examples include:

- Room application decisions.
- Room transfer decisions.

- Newly generated invoices.
- Debt reminders.
- Repair-status updates.
- Residence-declaration updates.
- Violation records.

13. Functional Requirement Traceability

Performed by: Trần Huỳnh Mạnh Đạt | **Reviewed by:** Đào Duy Anh | **Edited by:** Đào Duy Anh

Functional requirement	Related use cases	Coverage
FR01	UC-AUTH-01	Covered
FR02	UC-AUTH-02, UC-AUTH-03, UC-AUTH-04, UC-PRO-03	Covered
FR03	UC-PRO-01, UC-PRO-02, UC-STAY-01, UC-STAY-02, UC-STAY-03, UC-STU-01	Covered
FR04	UC-ADM-01	Covered
FR05	UC-ADM-02	Covered
FR06	UC-ADM-03	Covered
FR07	UC-ADM-04	Covered
FR08	-	Removed from scope
FR09	UC-ROOM-01	Covered
FR10	UC-STU-01	Covered
FR11	UC-ROOM-04, UC-ROOM-05, UC-ROOM-06	Covered

FR12	UC-ROOM-07	Covered
FR13	UC-ROOM-08, UC-ROOM-09, UC-ROOM-10	Covered
FR14	UC-CON-01, UC-CON-02, UC-CON-03, UC-CON-04	Covered
FR15	UC-CON-02	Covered
FR16	UC-CON-03	Covered
FR17	UC-CON-04, UC-CON-06	Covered
FR18	UC-CHK-01, UC-CHK-02	Covered
FR19	UC-CHK-02, UC-CHK-03	Covered — merged into checkout review/compensation, not a standalone use case (implemented: <code>Checkout.damages[]</code> , <code>/admin/checkouts</code> inspection modal)
FR20	UC-CHK-03	Covered
FR21	UC-CHK-04	Covered
FR22	UC-FIN-01, UC-FIN-02, UC-FIN-03, UC-FIN-07, UC-FIN-08, UC-FIN-09, UC-FIN-10, UC-FIN-11	Covered
FR23	UC-FIN-04, UC-FIN-05, UC-FIN-07	Covered
FR24	UC-FIN-06	Covered
FR25	UC-RES-01, UC-RES-05	Covered
FR26	UC-MNT-01, UC-MNT-03	Covered
FR27	UC-MNT-03	Covered

FR28	UC-MNT-02, UC-MNT-04, UC-MNT-05, UC-MNT-06, UC-MNT-07, UC-MNT-08, UC-MNT-09, UC-MNT-10	Covered
FR29	UC-COND-01, UC-COND-05	Covered
FR30	UC-COND-02, UC-COND-03, UC-COND-04	Covered

14. Student Feature Traceability

Performed by: Trần Huỳnh Mạnh Đạt | **Reviewed by:** Đào Duy Anh | **Edited by:** Đào Duy Anh

Student features	Related use cases
ST01-ST03 Personal profile	UC-PRO-01, UC-PRO-02, UC-PRO-03
ST04-ST06 Residence information	UC-STAY-01, UC-STAY-02, UC-STAY-03
ST07-ST10 Room rental	UC-ROOM-02, UC-ROOM-03, UC-ROOM-04, UC-ROOM-05
ST11-ST12 Contract	UC-CON-05, UC-CON-06
ST13-ST16 Invoice and payment	UC-FIN-08, UC-FIN-09, UC-FIN-10, UC-FIN-11
ST17-ST18 Repair requests	UC-MNT-01, UC-MNT-02
ST19-ST20 Feedback and suggestions	UC-FBK-01, UC-FBK-02
ST21-ST24 Residence declarations	UC-RES-01, UC-RES-02, UC-RES-03, UC-RES-04
ST25-ST26 Notifications	UC-NOT-01, UC-NOT-02
ST27-ST28 Evaluation and violations	UC-COND-04, UC-COND-05

15. Maintenance Staff Feature Traceability

Performed by: Trần Huỳnh Mạnh Đạt | **Reviewed by:** Đào Duy Anh | **Edited by:** Đào Duy Anh

Maintenance features	Related use cases
MT01	UC-MNT-05
MT02	UC-MNT-06
MT03-MT04	UC-MNT-07
MT05-MT06	UC-MNT-08
MT07	UC-MNT-09
MT08	UC-MNT-10

16. Floor Manager Function Reassignment (Diagram-Level Simplification)

Performed by: Trần Huỳnh Mạnh Đạt | **Reviewed by:** Đào Duy Anh | **Edited by:** Đào Duy Anh

Former Floor Manager functions	New responsible role and use cases
FM01-FM02 View students and residence information	Dormitory Manager - UC-STU-01
FM03 Track room conditions	Dormitory Manager - UC-ROOM-01
FM04-FM09 Manage meter readings	Dormitory Manager - UC-FIN-01
FM10-FM12 Track residence declarations	Dormitory Manager - UC-RES-05
FM13 Record violations	Dormitory Manager - UC-COND-01
FM14 Perform periodic evaluations	Dormitory Manager - UC-COND-03
FM15 Receive feedback	Dormitory Manager - UC-FBK-03
FM16 Track repair requests	Dormitory Manager - UC-MNT-04